Lab 06: Regression and Classification
Contribution Journals

ITAI 2373

Group: FA26_ML_6171_16347 5

Person 1: Kristina Akpan

Name: Kristina Akpan

Role: Person 1: Part 1, Regression vs. Classification, Knowledge Check 1, and final notebook assembly

For this lab I handled Part 1, the regression vs. classification section, and I wrote the answer to Knowledge Check question 1. I went through the notebook's explanation and the examples and then put the difference into plain words: regression predicts a number on a scale and classification predicts which group something falls in. I also wrote a short code cell for Part 1 so the idea wasn't just a definition. It loads the Titanic data and compares the two columns we predict later. Fare has 248 different values from 0 to about 512 dollars, so it's a regression target. Survived only has two values, 0 and 1, so it's a classification target. I added a quick check that labels each one and wrote up my observations under the output.

I also took on putting the final notebook together. When I first opened the file, all the code was squished onto one line in each cell, so nothing would run. I fixed the line breaks without changing any of the original text, then dropped everyone's code and answers into the right cells, ran the whole notebook top to bottom, and checked that every output showed and the answers were formatted the same way. Then I saved it with our naming format and got it ready to submit.

Person 2: Angel Martinez

Name: Angel Martinez

Role: Person 2: Part 2, Linear Regression setup and code (Task 1)

My responsibility was Task 1, the linear regression code. I created the LinearRegression model, fit it on X_train_reg and y_train_reg, generated predictions for the test set, and calculated the mean squared error. I kept the starter comments in place and wrote my code directly under each one, with a short inline comment explaining what each line does. I also printed lr_model.coef_ and the intercept, along with the feature list, so Kerry would have the coefficients in the correct order for the Knowledge Check.

While testing, I found two issues in the starter code. The final print statement called np.sqrt, but NumPy was never imported, so the cell raised an error until I added the import. The comment also claimed the model would be off by about 48 dollars, but the actual RMSE on this split was 58.01. Rather than copy the comment, I changed the print statement to calculate the dollar amount from the real result. Final output: MSE of 3364.92 and RMSE of 58.01.

Person 3: Kerry Walker

Name: Kerry Walker

Role: Person 3: Linear Regression interpretation and Knowledge Check 2 and 3

I was responsible for making sense of the regression output and answering Knowledge Check questions 2 and 3. Using the values Angel printed, I explained what each coefficient in lr_model.coef_ means: the Pclass slope of roughly negative 38.30 says each step down in class lowers the predicted fare by about 38 dollars, while the Age slope of roughly negative 0.44 says age barely matters at all. I also wrote the interpretation cell under the regression code, where I converted the MSE of 3364.92 back into dollars by taking the square root and compared that 58 dollar error to the average and median fares in the test set, which showed just how large the error really is.

For question 3, I explained why mean squared error fits a numeric target and accuracy fits a categorical one, and why accuracy would be close to meaningless for fares, since almost no prediction will match a real fare to the cent. I added the R squared value of about 0.19 to the interpretation because I felt the MSE alone didn't tell the full story about how little of the variation our two features explain.

Person 4: Thomas Jaymon

Name: Thomas Jaymon

Role: Person 4: Part 3, Logistic Regression setup and code (Task 2)

I took care of Task 2, the logistic regression code! I created the LogisticRegression model, trained it on the classification training data, made predictions on the test set, and calculated the accuracy score, which came out to 74.83%. Like Angel, I kept all the starter comments and added my code right under each one with a short note about what it does.

I also wanted Jamie to have more to work with than just one number, so I added a few extra lines of output. I printed how many passengers were in the test set (143), a baseline showing what you'd get by always guessing "did not survive" (60.84%), the confusion matrix, and the model's coefficients for Age, Pclass, and Sex. That way the interpretation could talk about where the model gets things wrong and which feature matters most, and not just whether 74.83% is good or bad.

Person 5: Jamie Lopez

Name: Jamie Lopez

Role: Person 5: Logistic Regression interpretation and final reflection

My part of the lab was interpreting the classification results and writing the group reflection at the end of the notebook. Using the output Thomas set up, I explained what the 74.83% accuracy actually means in terms of people: 107 of the 143 test passengers were predicted correctly and 36 were not. I compared that to the 60.84% baseline to show the model learned something real, walked through the confusion matrix to show that most mistakes were survivors the model missed, and connected the coefficients to what we know about the Titanic, where sex and passenger class shaped who survived.

I then wrote the reflection cell, which covers what we learned, the challenges we ran into (the missing NumPy import and the 48 versus 58 dollar mismatch), our main insights, and how this lab connects to our earlier NLP work, where we turned text into features that models like this one can classify.